CORE PURE / VECTORS, MATRICES & LINEAR TRANSFORMATIONS

Matrix Determinants & Inverses

Worked solutions 10 questions · 78 marks

EXAM BOARDS Edexcel 9FM0 · AQA 7367 · OCR A H245 · OCR MEI H645 · CIE 9231

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1.

$$\mathbf{B} = \begin{pmatrix} 1 & a \\ 1 & 3 \end{pmatrix}$$

where a is a real constant and $a \neq 3$.

(a) Find $\mathbf{B}^{-1}$ in terms of a. **[2]**

Given that

$$\mathbf{B}^{-1} = 2\mathbf{I} - \frac{1}{2}\mathbf{B}$$

where $\mathbf{I}$ is the 2×2 identity matrix,

(b) find the value of a. **[3]**

Solution

(a) For

$$\mathbf{B} = \begin{pmatrix} 1 & a \\ 1 & 3 \end{pmatrix}$$

the determinant is

$$\det(\mathbf{B}) = 1 \times 3 - 1 \times a = 3 - a$$

Since $a \neq 3$, $\det(\mathbf{B}) \neq 0$, so $\mathbf{B}^{-1}$ exists.

So

$$\mathbf{B}^{-1} = \frac{1}{3-a}\begin{pmatrix} 3 & -a \\ -1 & 1 \end{pmatrix}$$

(b) We are given

$$\mathbf{B}^{-1} = 2\mathbf{I} - \frac{1}{2}\mathbf{B}$$

Now

$$2\mathbf{I} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \qquad \frac{1}{2}\mathbf{B} = \begin{pmatrix} \frac{1}{2} & \frac{a}{2} \\ \frac{1}{2} & \frac{3}{2} \end{pmatrix}$$

So

$$2\mathbf{I} - \frac{1}{2}\mathbf{B} = \begin{pmatrix} 2 - \frac{1}{2} & 0 - \frac{a}{2} \\ 0 - \frac{1}{2} & 2 - \frac{3}{2} \end{pmatrix} = \begin{pmatrix} \frac{3}{2} & -\frac{a}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 3 & -a \\ -1 & 1 \end{pmatrix}$$

From part (a):

$$\mathbf{B}^{-1} = \frac{1}{3-a}\begin{pmatrix} 3 & -a \\ -1 & 1 \end{pmatrix}$$

Since this is equal to

$$\frac{1}{2}\begin{pmatrix} 3 & -a \\ -1 & 1 \end{pmatrix}$$

the scalar factors must be equal:

$$\frac{1}{3-a} = \frac{1}{2}$$

Hence

$$3 - a = 2$$

and therefore

$$a = 1$$

2. Given that

$$\mathbf{A} = \begin{pmatrix} k+1 & 2 \\ 1-2k & k-4 \\ 2k-1 & 4-k \end{pmatrix} \quad \text{and} \quad \mathbf{B} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$$

where k is a constant,

(a) determine the matrix $\mathbf{BA}$. **[2]**

(b) hence find the values of k for which $\mathbf{BA}$ is singular. **[4]**

Solution

(a) Since $\mathbf{B}$ is 2×3 and $\mathbf{A}$ is 3×2, the product $\mathbf{BA}$ is a 2×2 matrix.

Multiply rows of $\mathbf{B}$ by columns of $\mathbf{A}$:

$$\mathbf{BA} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} k+1 & 2 \\ 1-2k & k-4 \\ 2k-1 & 4-k \end{pmatrix}$$

$$\begin{aligned} (1,1) \text{ entry} &= (1)(k+1) + (1)(1-2k) + (1)(2k-1) = k+1 \\ (1,2) \text{ entry} &= (1)(2) + (1)(k-4) + (1)(4-k) = 2 \\ (2,1) \text{ entry} &= (1)(k+1) + (0)(1-2k) + (1)(2k-1) = 3k \\ (2,2) \text{ entry} &= (1)(2) + (0)(k-4) + (1)(4-k) = 6-k \end{aligned}$$

Therefore

$$\mathbf{BA} = \begin{pmatrix} k+1 & 2 \\ 3k & 6-k \end{pmatrix}$$

(b) A matrix is singular if its determinant is 0.

So

$$\det(\mathbf{BA}) = \begin{vmatrix} k+1 & 2 \\ 3k & 6-k \end{vmatrix} = (k+1)(6-k) - 2(3k)$$

$$\begin{aligned} \det(\mathbf{BA}) &= 6k + 6 - k^2 - k - 6k \\ &= -k^2 - k + 6 \\ &= -(k^2 + k - 6) \\ &= -(k+3)(k-2) \end{aligned}$$

So $\mathbf{BA}$ is singular when

$$-(k+3)(k-2) = 0$$

Hence

$$k = -3 \quad \text{or} \quad k = 2$$

3. A company manufactures only three models of bicycle: Commuter, Electric and Folding.

Each bicycle manufactured is one of these three models.

In 2023, a total of 1000 bicycles were manufactured.

The number of Folding bicycles manufactured was 200 fewer than the combined number of Commuter and Electric bicycles.

In 2024 the number manufactured of

- Commuter increased by 10%
- Electric decreased by 5%
- Folding increased by 5%

In 2024, the total number of bicycles manufactured increased by 3.8%.

(a) (i) Define, for each model, a variable for the number of bicycles manufactured in 2023. **[2]**

(ii) Using your variables from part (a)(i), write down three equations that model this situation. **[4]**

(b) By forming and solving a matrix equation, determine how many bicycles of each model were manufactured in 2023. **[4]**

Solution

(a) (i) Let

$$C = \text{number of Commuter bicycles made in 2023}$$

$$E = \text{number of Electric bicycles made in 2023}$$

$$F = \text{number of Folding bicycles made in 2023}$$

(ii) In 2023, the total number manufactured was 1000, so

$$C + E + F = 1000$$

The number of Folding bicycles was 200 fewer than the combined number of Commuter and Electric bicycles, so

$$F = C + E - 200$$

In 2024, the total increased by 3.8%, so the new total was

$$1000 \times 1.038 = 1038$$

Using the percentage changes for each model gives

$$1.10C + 0.95E + 1.05F = 1038$$

So the three equations are

$$C + E + F = 1000, \qquad F = C + E - 200, \qquad 1.10C + 0.95E + 1.05F = 1038$$

(b) Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ -1 & -1 & 1 \\ 1.10 & 0.95 & 1.05 \end{pmatrix}, \qquad \mathbf{x} = \begin{pmatrix} C \\ E \\ F \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 1000 \\ -200 \\ 1038 \end{pmatrix}$$

Then the matrix equation is

$$\mathbf{Ax} = \mathbf{b}$$

Using a calculator, the inverse is

$$\mathbf{A}^{-1} = \begin{pmatrix} 1 & 1 & 1 \\ -1 & -1 & 1 \\ 1.10 & 0.95 & 1.05 \end{pmatrix}^{-1} = \begin{pmatrix} -\frac{20}{3} & -\frac{1}{3} & \frac{20}{3} \\ \frac{43}{6} & -\frac{1}{6} & -\frac{20}{3} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}$$

So

$$\begin{aligned} \begin{pmatrix} C \\ E \\ F \end{pmatrix} &= \begin{pmatrix} 1 & 1 & 1 \\ -1 & -1 & 1 \\ 1.10 & 0.95 & 1.05 \end{pmatrix}^{-1} \begin{pmatrix} 1000 \\ -200 \\ 1038 \end{pmatrix} \\ &= \begin{pmatrix} -\frac{20}{3} & -\frac{1}{3} & \frac{20}{3} \\ \frac{43}{6} & -\frac{1}{6} & -\frac{20}{3} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix} \begin{pmatrix} 1000 \\ -200 \\ 1038 \end{pmatrix} \\ &= \begin{pmatrix} 320 \\ 280 \\ 400 \end{pmatrix} \end{aligned}$$

So

$$\text{Commuter } = 320, \qquad \text{Electric } = 280, \qquad \text{Folding } = 400$$

4. A community centre offers three evening courses: art, dance and pottery. At the start of the year there were 580 enrolments in total. The number enrolled in art was 20 more than the total number enrolled in dance and pottery combined.

By the end of the year:

- the number enrolled in art had decreased by 5%
- the number enrolled in dance had increased by 10%
- the number enrolled in pottery had increased by 15%
- overall enrolment had increased by 20

Form and solve a matrix equation to find the initial number enrolled in each course. **[7]**

Solution

Let the initial numbers enrolled in art, dance and pottery be A, D and P respectively.

From the information given:

$$A + D + P = 580$$

Art was 20 more than dance and pottery combined, so

$$A = D + P + 20$$

which gives

$$A - D - P = 20$$

By the end of the year, the total enrolment had increased by 20, so the new total was 600. Therefore

$$0.95A + 1.10D + 1.15P = 600$$

To avoid decimals, multiply this last equation by 20:

$$19A + 22D + 23P = 12\,000$$

So the matrix equation is

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ 19 & 22 & 23 \end{pmatrix} \begin{pmatrix} A \\ D \\ P \end{pmatrix} = \begin{pmatrix} 580 \\ 20 \\ 12\,000 \end{pmatrix}$$

Using a calculator, the inverse is

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ 19 & 22 & 23 \end{pmatrix}^{-1} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 21 & -2 & -1 \\ -\frac{41}{2} & \frac{3}{2} & 1 \end{pmatrix}$$

So

$$\begin{pmatrix} A \\ D \\ P \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ 19 & 22 & 23 \end{pmatrix}^{-1} \begin{pmatrix} 580 \\ 20 \\ 12\,000 \end{pmatrix}$$
$$= \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 21 & -2 & -1 \\ -\frac{41}{2} & \frac{3}{2} & 1 \end{pmatrix} \begin{pmatrix} 580 \\ 20 \\ 12\,000 \end{pmatrix}$$
$$= \begin{pmatrix} 300 \\ 140 \\ 140 \end{pmatrix}$$

So

$$\text{Art } = 300, \qquad \text{Dance } = 140, \qquad \text{Pottery } = 140$$

5. The matrix $\mathbf{A}$ is given by

$$\mathbf{A} = \begin{pmatrix} 2 & k & 1 \\ 1 & 0 & 2 \\ k & -1 & 1 \end{pmatrix}$$

where k is a real constant.

(a) Show that $\mathbf{A}$ is non-singular for all real values of k. **[3]**

(b) Given that

$$\mathbf{A}^{-1} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{4} & -\frac{3}{4} \\ -\frac{1}{4} & \frac{3}{4} & -\frac{1}{4} \end{pmatrix}$$

find the value of k. **[2]**

Solution

(a) Expanding $\det(\mathbf{A})$ along the first row:

$$\begin{aligned} \det(\mathbf{A}) &= 2\begin{vmatrix} 0 & 2 \\ -1 & 1 \end{vmatrix} - k\begin{vmatrix} 1 & 2 \\ k & 1 \end{vmatrix} + 1\begin{vmatrix} 1 & 0 \\ k & -1 \end{vmatrix} \\ &= 2(0 \times 1 - 2(-1)) - k(1 \times 1 - 2k) + 1(1 \times (-1) - 0 \times k) \\ &= 2(2) - k(1 - 2k) - 1 \\ &= 4 - k + 2k^2 - 1 \\ &= 2k^2 - k + 3 \end{aligned}$$

Now consider the quadratic $2k^2 - k + 3$. Its discriminant is

$$(-1)^2 - 4(2)(3) = 1 - 24 = -23 < 0$$

and the coefficient of k^2 is positive, so

$$2k^2 - k + 3 > 0 \quad \text{for all } k \in \mathbb{R}$$

Hence

$$\det(\mathbf{A}) \neq 0 \quad \text{for all real } k$$

Therefore $\mathbf{A}$ is non-singular for all real values of k.

(b) Let the given matrix be $\mathbf{B}$. Since $\mathbf{B} = \mathbf{A}^{-1}$, we must have

$$\mathbf{AB} = \mathbf{I}$$

Using the $(1,1)$ entry of $\mathbf{AB}$:

$$\begin{pmatrix} 2 & k & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{2} \\ \frac{1}{4} \\ -\frac{1}{4} \end{pmatrix} = 1$$

$$2 \times \frac{1}{2} + k \times \frac{1}{4} + 1 \times \left(-\frac{1}{4}\right) = 1$$

$$1 + \frac{k}{4} - \frac{1}{4} = 1$$

$$\frac{k+3}{4} = 1$$

$$k + 3 = 4$$

$$k = 1$$

6. A cinema chain sells three types of annual pass, A, B and C.

It is known that pass A makes a profit of £16 per pass sold, pass B makes a profit of £15 per pass sold and pass C makes a profit of £12 per pass sold.

Between 2023 and 2024

- sales of pass A increased by 25%
- sales of pass B decreased by 20%
- sales of pass C increased by 50%

In total 900 000 passes were sold in 2023.

In total 1 050 000 passes were sold in 2024.

The total profit from these passes in 2024 was £15.12 million.

Form and solve a matrix equation to find, to 2 significant figures, the number of each type of pass sold in 2023. **[8]**

Solution

Let the numbers of passes of types A, B and C sold in 2023 be x, y and z respectively.

Then the numbers sold in 2024 were:

$$1.25x, \quad 0.8y, \quad 1.5z$$

In total 900 000 passes were sold in 2023, so

$$x + y + z = 900\,000$$

In total 1 050 000 passes were sold in 2024, so

$$1.25x + 0.8y + 1.5z = 1\,050\,000$$

To avoid decimals, multiply this equation by 20:

$$25x + 16y + 30z = 21\,000\,000$$

Now use the 2024 profit information.

Profit per pass is:

$$A : £16, \qquad B : £15, \qquad C : £12$$

So the total profit in 2024 is

$$16(1.25x) + 15(0.8y) + 12(1.5z) = 15\,120\,000$$

This simplifies to

$$20x + 12y + 18z = 15\,120\,000$$

So the matrix equation is

$$\begin{pmatrix} 1 & 1 & 1 \\ 25 & 16 & 30 \\ 20 & 12 & 18 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 900\,000 \\ 21\,000\,000 \\ 15\,120\,000 \end{pmatrix}$$

Let

$$\mathbf{M} = \begin{pmatrix} 1 & 1 & 1 \\ 25 & 16 & 30 \\ 20 & 12 & 18 \end{pmatrix}$$

Expanding along the first row:

$$\begin{aligned} \det(\mathbf{M}) &= 1\begin{vmatrix} 16 & 30 \\ 12 & 18 \end{vmatrix} - 1\begin{vmatrix} 25 & 30 \\ 20 & 18 \end{vmatrix} + 1\begin{vmatrix} 25 & 16 \\ 20 & 12 \end{vmatrix} \\ &= (-72) - (-150) + (-20) \\ &= 58 \end{aligned}$$

Since $\det(\mathbf{M}) \neq 0$, the inverse exists and

$$\mathbf{M}^{-1} = \frac{1}{\det(\mathbf{M})}\,\mathrm{adj}(\mathbf{M})$$

The matrix of cofactors is

$$\begin{pmatrix} -72 & 150 & -20 \\ -6 & -2 & 8 \\ 14 & -5 & -9 \end{pmatrix}$$

The adjugate, $\mathrm{adj}(\mathbf{M})$, is the transposed matrix of cofactors:

$$\mathrm{adj}(\mathbf{M}) = \begin{pmatrix} -72 & -6 & 14 \\ 150 & -2 & -5 \\ -20 & 8 & -9 \end{pmatrix}$$

Hence

$$\mathbf{M}^{-1} = \frac{1}{58}\begin{pmatrix} -72 & -6 & 14 \\ 150 & -2 & -5 \\ -20 & 8 & -9 \end{pmatrix}$$

Now multiply both sides of the matrix equation by $\mathbf{M}^{-1}$:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{1}{58}\begin{pmatrix} -72 & -6 & 14 \\ 150 & -2 & -5 \\ -20 & 8 & -9 \end{pmatrix}\begin{pmatrix} 900\,000 \\ 21\,000\,000 \\ 15\,120\,000 \end{pmatrix}$$

This gives

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{1}{58}\begin{pmatrix} 20\,880\,000 \\ 17\,400\,000 \\ 13\,920\,000 \end{pmatrix} = \begin{pmatrix} 360\,000 \\ 300\,000 \\ 240\,000 \end{pmatrix}$$

So the numbers of passes of each type sold in 2023 were

$$A:\ 3.6 \times 10^5, \qquad B:\ 3.0 \times 10^5, \qquad C:\ 2.4 \times 10^5 \qquad \text{(2 s.f.)}$$

7.

$$\mathbf{A} = \begin{pmatrix} a & 1 & 0 \\ 2 & 3 & 1 \\ 0 & 1 & a \end{pmatrix}$$

where a is a real constant.

(a) Determine the values of a for which $\mathbf{A}$ is singular. **[2]**

(b) Hence, for all other values of a, find $\mathbf{A}^{-1}$ in terms of a. **[4]**

Solution

(a) Expand $\det(\mathbf{A})$ along the first row:

$$\begin{aligned} \det(\mathbf{A}) &= a\begin{vmatrix} 3 & 1 \\ 1 & a \end{vmatrix} - 1\begin{vmatrix} 2 & 1 \\ 0 & a \end{vmatrix} + 0\begin{vmatrix} 2 & 3 \\ 0 & 1 \end{vmatrix} \\ &= a(3a-1) - 2a \\ &= 3a^2 - 3a \\ &= 3a(a-1) \end{aligned}$$

$\mathbf{A}$ is singular when $\det(\mathbf{A}) = 0$, so

$$3a(a-1) = 0 \quad \Longrightarrow \quad a = 0 \text{ or } a = 1$$

(b) For $a \neq 0, 1$, the inverse exists and

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})}\operatorname{adj}(\mathbf{A})$$

From part (a):

$$\det(\mathbf{A}) = 3a(a-1)$$

The matrix of cofactors is

$$\begin{pmatrix} 3a-1 & -2a & 2 \\ -a & a^2 & -a \\ 1 & -a & 3a-2 \end{pmatrix}$$

The adjugate, $\operatorname{adj}(\mathbf{A})$, is the transposed matrix of cofactors:

$$\operatorname{adj}(\mathbf{A}) = \begin{pmatrix} 3a-1 & -a & 1 \\ -2a & a^2 & -a \\ 2 & -a & 3a-2 \end{pmatrix}$$

Therefore

$$\mathbf{A}^{-1} = \frac{1}{3a(a-1)}\begin{pmatrix} 3a-1 & -a & 1 \\ -2a & a^2 & -a \\ 2 & -a & 3a-2 \end{pmatrix} \qquad (a \neq 0, 1)$$

8.

$$\mathbf{A} = \begin{pmatrix} k & 2 & 1 \\ -1 & 3 & 0 \\ 4 & -2 & 1 \end{pmatrix} \qquad \mathbf{B} = \begin{pmatrix} 3 & -4 & -3 \\ 1 & k-4 & -1 \\ -10 & 2k+8 & 3k+2 \end{pmatrix}$$

where k is a constant.

(a) Determine the value of the constant c for which

$$\mathbf{AB} = (3k - c)\mathbf{I}$$ **[2]**

(b) Hence determine the value of k for which $\mathbf{A}^{-1}$ does not exist. **[2]**

Given that $\mathbf{A}^{-1}$ does exist,

(c) write down $\mathbf{A}^{-1}$ in terms of k. **[1]**

(d) Use the answer to part (c) to solve the simultaneous equations

$$\begin{aligned} kx + 2y + z &= 1 \\ -x + 3y &= 2 \\ 4x - 2y + z &= 3 \end{aligned}$$

giving the values of x, y and z in simplest form in terms of k. **[3]**

Solution

(a) Multiplying $\mathbf{A}$ by $\mathbf{B}$:

$$\begin{aligned} \mathbf{AB} &= \begin{pmatrix} k & 2 & 1 \\ -1 & 3 & 0 \\ 4 & -2 & 1 \end{pmatrix} \begin{pmatrix} 3 & -4 & -3 \\ 1 & k-4 & -1 \\ -10 & 2k+8 & 3k+2 \end{pmatrix} \\ &= \begin{pmatrix} 3k+2-10 & -4k+2(k-4)+(2k+8) & -3k+2(-1)+(3k+2) \\ -3+3 & 4+3(k-4) & 3-3 \\ 12-2-10 & -16-2(k-4)+(2k+8) & -12+2+(3k+2) \end{pmatrix} \\ &= \begin{pmatrix} 3k-8 & 0 & 0 \\ 0 & 3k-8 & 0 \\ 0 & 0 & 3k-8 \end{pmatrix} = (3k-8)\mathbf{I} \end{aligned}$$

Comparing with $(3k - c)\mathbf{I}$, we get $c = 8$.

(b) From part (a):

$$\mathbf{AB} = (3k - 8)\mathbf{I}$$

If $3k - 8 \neq 0$, then

$$\mathbf{A}\left(\frac{1}{3k-8}\mathbf{B}\right) = \mathbf{I}$$

so $\mathbf{A}^{-1}$ exists.

Therefore $\mathbf{A}^{-1}$ does not exist when

$$3k - 8 = 0$$

Hence $k = \frac{8}{3}$.

(c) Given that $\mathbf{A}^{-1}$ exists, $k \neq \frac{8}{3}$, so from part (a)

$$\mathbf{A}^{-1} = \frac{1}{3k-8}\mathbf{B} = \frac{1}{3k-8}\begin{pmatrix} 3 & -4 & -3 \\ 1 & k-4 & -1 \\ -10 & 2k+8 & 3k+2 \end{pmatrix}$$

(d) Write the equations as

$$\mathbf{A}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

So

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \mathbf{A}^{-1}\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \frac{1}{3k-8}\begin{pmatrix} 3 & -4 & -3 \\ 1 & k-4 & -1 \\ -10 & 2k+8 & 3k+2 \end{pmatrix}\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

Now multiply the matrices:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{1}{3k-8}\begin{pmatrix} 3-8-9 \\ 1+2(k-4)-3 \\ -10+2(2k+8)+3(3k+2) \end{pmatrix} = \frac{1}{3k-8}\begin{pmatrix} -14 \\ 2k-10 \\ 13k+12 \end{pmatrix}$$

Hence

$$x = -\frac{14}{3k-8}, \qquad y = \frac{2k-10}{3k-8}, \qquad z = \frac{13k+12}{3k-8}$$

9. You are given that a is a real parameter.

The matrix $\mathbf{A}$ is given by

$$\mathbf{A} = \begin{pmatrix} 1 & -2 & 3 \\ 4 & 1 & -1 \\ a & 0 & 1 \end{pmatrix}$$

(a) Find an expression for $\det(\mathbf{A})$ in terms of a. **[2]**

You are given the following system of equations in x, y and z.

$$\begin{aligned} x - 2y + 3z &= -6 \\ 4x + y - z &= 7 \\ ax + z &= a - 1 \end{aligned}$$

The system can be written in the form

$$\mathbf{A}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -6 \\ 7 \\ a-1 \end{pmatrix}$$

(b) (i) In the case where $\mathbf{A}$ is not singular, solve the given system by using $\mathbf{A}^{-1}$. **[6]**

(ii) In the case where $\mathbf{A}$ is singular, describe the configuration of the planes whose equations are the three equations of the system. **[3]**

The transformation represented by $\mathbf{A}$ is denoted by T.
A solid of volume 12 is transformed by T to an image in 3-D space.

(c) (i) Determine the range of values of a for which the orientation of the image is the reverse of the orientation of the object. **[1]**

(ii) Determine the range of values of a for which the volume of the image is less than the volume of the object. **[2]**

Solution

(a) Expanding along the first row:

$$\begin{aligned} \det(\mathbf{A}) &= 1\begin{vmatrix} 1 & -1 \\ 0 & 1 \end{vmatrix} - (-2)\begin{vmatrix} 4 & -1 \\ a & 1 \end{vmatrix} + 3\begin{vmatrix} 4 & 1 \\ a & 0 \end{vmatrix} \\ &= 1(1) + 2(4 + a) + 3(0 - a) \\ &= 1 + 8 + 2a - 3a \\ &= 9 - a \end{aligned}$$

(b) (i) From part (a), $\mathbf{A}$ is non-singular when

$$9 - a \neq 0 \quad \Longrightarrow \quad a \neq 9$$

For $a \neq 9$, the inverse exists and

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \operatorname{adj}(\mathbf{A})$$

The matrix of cofactors is

$$\begin{pmatrix} 1 & -a-4 & -a \\ 2 & 1-3a & -2a \\ -1 & 13 & 9 \end{pmatrix}$$

The adjugate, $\operatorname{adj}(\mathbf{A})$, is the transposed matrix of cofactors:

$$\operatorname{adj}(\mathbf{A}) = \begin{pmatrix} 1 & 2 & -1 \\ -a-4 & 1-3a & 13 \\ -a & -2a & 9 \end{pmatrix}$$

Hence

$$\mathbf{A}^{-1} = \frac{1}{9-a} \begin{pmatrix} 1 & 2 & -1 \\ -a-4 & 1-3a & 13 \\ -a & -2a & 9 \end{pmatrix}$$

Therefore

$$\begin{aligned} \begin{pmatrix} x \\ y \\ z \end{pmatrix} &= \mathbf{A}^{-1} \begin{pmatrix} -6 \\ 7 \\ a-1 \end{pmatrix} \\ &= \frac{1}{9-a} \begin{pmatrix} 1 & 2 & -1 \\ -a-4 & 1-3a & 13 \\ -a & -2a & 9 \end{pmatrix} \begin{pmatrix} -6 \\ 7 \\ a-1 \end{pmatrix} \\ &= \frac{1}{9-a} \begin{pmatrix} -6+14-(a-1) \\ (-a-4)(-6)+(1-3a)7+13(a-1) \\ (-a)(-6)+(-2a)7+9(a-1) \end{pmatrix} \\ &= \frac{1}{9-a} \begin{pmatrix} 9-a \\ 18-2a \\ a-9 \end{pmatrix} \\ &= \frac{1}{9-a} \begin{pmatrix} 9-a \\ 2(9-a) \\ -(9-a) \end{pmatrix} \\ &= \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \end{aligned}$$

So, for $a \neq 9$, the solution is

$$x = 1, \qquad y = 2, \qquad z = -1$$

(ii) $\mathbf{A}$ is singular when

$$9 - a = 0 \quad \Longrightarrow \quad a = 9$$

Then the three plane equations are

$$x - 2y + 3z = -6$$

$$4x + y - z = 7$$

$$9x + z = 8$$

Now

$$\begin{pmatrix} 9 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} + 2\begin{pmatrix} 4 \\ 1 \\ -1 \end{pmatrix}$$

and

$$8 = -6 + 2 \times 7$$

So the third equation is obtained by adding the first equation to 2 times the second equation.

Therefore any point which lies on the intersection of the first two planes also lies on the third plane. Since the first two planes are distinct and not parallel, they intersect in a line.

Hence, when $a = 9$, the three planes form a **sheaf**: they are distinct and share a common line of intersection, so there are infinitely many solutions.

(c) (i) The orientation is reversed when $\det(\mathbf{A}) < 0$, so

$$9 - a < 0$$

$$a > 9$$

(ii) The volume scale factor is

$$|\det(\mathbf{A})| = |9 - a|$$

So the image volume is

$$12|9 - a|$$

We want this to be less than 12, so

$$12|9 - a| < 12$$

$$|9 - a| < 1$$

Therefore

$$-1 < 9 - a < 1$$

$$8 < a < 10$$

10. Three planes have equations

$$\begin{aligned} x + 2y - z &= 4 \\ 3x + ky + 2z &= k + 11 \\ ky + 4z &= k + 4 \end{aligned}$$

where k is a constant.

(a) Show that the planes have a unique point of intersection except for one value of k, which should be found. **[4]**

(b) Show that, whenever the planes have a unique point of intersection, this point is independent of k. Find the point. **[5]**

Solution

(a) Write the equations in matrix form as

$$\mathbf{A}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ k+11 \\ k+4 \end{pmatrix}, \qquad \mathbf{A} = \begin{pmatrix} 1 & 2 & -1 \\ 3 & k & 2 \\ 0 & k & 4 \end{pmatrix}$$

The three planes have a unique point of intersection exactly when this linear system has a unique solution, that is when

$$\det(\mathbf{A}) \neq 0$$

Now expand the determinant along the first row:

$$\begin{aligned} \det(\mathbf{A}) &= 1\begin{vmatrix} k & 2 \\ k & 4 \end{vmatrix} - 2\begin{vmatrix} 3 & 2 \\ 0 & 4 \end{vmatrix} + (-1)\begin{vmatrix} 3 & k \\ 0 & k \end{vmatrix} \\ &= 1(4k - 2k) - 2(12) + (-1)(3k) \\ &= 2k - 24 - 3k \\ &= -(k + 24) \end{aligned}$$

So

$$\det(\mathbf{A}) \neq 0 \iff -(k+24) \neq 0 \iff k \neq -24$$

Hence the planes have a unique point of intersection for all values of k except

$$k = -24$$

(b) To find the intersection point, eliminate the ky term first.

Subtracting the third equation from the second:

$$(3x + ky + 2z) - (ky + 4z) = (k + 11) - (k + 4)$$

$$3x - 2z = 7$$

Now multiply the first equation by 3:

$$3x + 6y - 3z = 12$$

Subtract $3x - 2z = 7$ from this:

$$(3x + 6y - 3z) - (3x - 2z) = 12 - 7$$
$$6y - z = 5$$

so

$$z = 6y - 5$$

Substitute this into the first equation:

$$x + 2y - (6y - 5) = 4$$
$$x - 4y + 5 = 4$$
$$x = 4y - 1$$

Now substitute $z = 6y - 5$ into the third equation:

$$ky + 4(6y - 5) = k + 4$$
$$ky + 24y - 20 = k + 4$$
$$(k + 24)y = k + 24$$

Whenever there is a unique point of intersection, part (a) shows that $k \neq -24$. So we may divide by $k + 24$:

$$y = 1$$

Then

$$z = 6(1) - 5 = 1$$

and

$$x = 4(1) - 1 = 3$$

Therefore, whenever the planes have a unique common point, it is

$$(3, 1, 1)$$

This point is independent of k.